

1 nested quantifiers

$$\forall x \exists y \forall z [(x + y) \cdot z = 0]$$

translation: for all x , there exists a y such that for all z , the result of x plus y , multiplied by z , is zero.

1.1 nested quantifier order

the order of the quantifiers matter! when reading from left to right, *later* quantifiers are within the scope of the *earlier* ones.

$$\begin{array}{ll} \forall x \exists y (x + y = 0) & \checkmark \\ \exists y \forall x (x + y = 0) & \times \end{array}$$

1.2 translation

yes, we can (obviously) still formalize sentences from english. let's see this example:

every real number except zero has a multiplicative inverse.

we can rewrite that into a sentence that is easier for us to translate:

for **every** real number x , if $x \neq 0$,
then there **exists** a real number y such that $x \cdot y = 1$.

finally, we can formalize that sentence:

$$\forall x [(x \neq 0) \rightarrow \exists y [x \cdot y = 1]]$$